

Bayesian Estimation of Allele Frequencies Across Human Populations: Detecting Signatures of Natural Selection Using the Beta-Binomial Model

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May 2026

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1 Abstract

Natural selection leaves detectable signatures in the human genome by shifting allele frequencies away from neutral expectations. Classical methods for detecting selection treat allele frequencies as fixed point estimates, ignoring the uncertainty inherent in finite sample sizes. This project applies a Bayesian Beta-Binomial model to allele count data from the 1000 Genomes Project Phase 3, covering 2,504 individuals across five continental superpopulations (AFR, AMR, EAS, EUR, SAS). We analyzed 12 variants at loci documented to be under positive or balancing selection alongside a 10-SNP neutral background panel. For each variant-population combination, we derived analytically tractable Beta posteriors via the conjugate update rule and quantified inter-population differentiation using pairwise Bhattacharyya distances (BD). A one-sided Wilcoxon rank-sum test confirmed that selection loci show significantly greater population differentiation than background loci ($W = 99$, $p < 0.001$). Seventeen of the 18 possible pairwise comparisons for the four core validation loci (LCT, DARC, EDAR, SLC24A5) yielded BD scores more than two standard deviations above the background mean. The sickle-cell locus (HBB) showed a distinctly lower BD, consistent with balancing rather than directional selection. Haplotype-based linkage disequilibrium analysis corroborated these findings: positive selection loci exhibited substantially more extended haplotype blocks than background variants. Together, these results validate the Bayesian Beta-Binomial approach as a computationally lightweight yet statistically rigorous framework for genomic population differentiation analysis.

2 Introduction

2.1 Background and Significance

The human genome carries a record of adaptive evolution: where natural selection has acted on standing variation or new mutations, it distorts the population-genetic landscape in ways that are measurable tens of thousands of years after the selective pressure operated. Identifying the genomic loci that have experienced selection is fundamental to understanding human phenotypic diversity, population-specific disease susceptibility, and the evolutionary history of our species.

The most common quantitative measure of inter-population allele frequency differentiation is Wright’s F_{ST} , which summarizes variation in allele frequencies across populations as a fraction of total heterozygosity (Wright, 1951). Although powerful and widely used, F_{ST} and its derivatives treat the observed allele frequency as a fixed, exact quantity. In practice, that frequency is estimated from a finite sample of chromosomes, and the estimate carries sampling uncertainty that is not reflected in the point value. This matters especially when populations are sampled at different depths: a variant observed at 3/10 chromosomes and one observed at 300/1,000 chromosomes both yield a point estimate of 0.30, yet carry very different amounts of information. Naive frequency comparisons that ignore this uncertainty can misrank loci when sample sizes vary substantially.

A Bayesian framework addresses this limitation directly. By treating allele frequency as a random variable and constructing a posterior distribution over its value given the observed counts, we obtain an estimate that explicitly encodes uncertainty and naturally regularizes extreme values arising from small samples. The Beta-Binomial model is the natural conjugate choice: a Beta prior on frequency combined with a Binomial likelihood for allele counts yields a Beta posterior in closed form, with no need for numerical sampling (Gelman et al., 2013). Downstream comparisons between populations can then be made at the level of complete posterior distributions rather than at the level of scalar point estimates, yielding richer and more calibrated inferences about differentiation.

Beyond improved uncertainty quantification, the Bayesian framework produces full posterior distributions that support direct probabilistic statements — for example, the probability that allele frequency in population A exceeds that in population B — and enables the use of information-theoretic divergence measures such as the Kullback-Leibler divergence and Bhattacharyya distance to characterize distributional overlap. These metrics capture not only the difference in location (analogous to F_{ST}) but also the difference in uncertainty, rewarding well-resolved comparisons and penalizing those where posterior distributions are diffuse.

2.2 Biological Question

Can a Bayesian Beta-Binomial model, using allele count data from the 1000 Genomes Project, identify SNP loci with significantly elevated inter-population posterior differentiation, thereby recovering known signals of natural selection against a neutral background?

2.3 Validation Loci

We anchor the analysis to four well-characterized loci whose selection histories have been established by independent lines of evidence (Figure and Table in Section 3.2):

LCT (rs4988235) — The derived A allele of this variant in the MCM6 gene regulatory region maintains lactase expression into adulthood (lactase persistence). It has experienced one of the strongest selective sweeps in European prehistory, reaching ~74% frequency in Northern Europeans

while remaining near-zero in African and East Asian populations (Tishkoff et al., 2007; Bersaglieri et al., 2004). Published iHS scores exceed 5.8 in the CEU population.

DARC/ACKR1 (rs2814778) — The T allele abolishes expression of the Duffy antigen receptor for chemokines on red blood cells (Duffy-null phenotype). Because *Plasmodium vivax* malaria uses the Duffy antigen as its obligate invasion receptor, Duffy-null individuals are highly resistant to vivax malaria. This variant has been driven to near-fixation (~97%) in sub-Saharan African populations while remaining rare elsewhere (Hamblin and Di Rienzo, 2000).

EDAR (rs3827760) — The derived A allele of the ectodysplasin A receptor gene increases hair shaft diameter, sweat gland density, and breast size, and is at high frequency in East Asian and Native American populations. Haplotype analyses (XP-EHH) place this among the strongest sweep signals in EAS genomes, with iHS scores around 6.2 in CHB (Fujimoto et al., 2008).

SLC24A5 (rs1426654) — The A allele at this solute carrier locus is a major determinant of reduced skin pigmentation and has risen to near-fixation in European and South Asian populations while remaining rare in Africans and East Asians (Lamason et al., 2005). iHS in CEU exceeds 5.5.

In addition to these four validation loci, we expand the analysis to eight further variants documented in the literature: HBB (rs334, sickle-cell, balancing selection in AFR), ABCC11 (rs17822931, dry earwax, positive selection in EAS), SLC45A2 (rs16891982, pigmentation, EUR), HERC2/OCA2 (rs12913832, blue eye color, EUR), MC1R (rs1805007, red hair, EUR), ADH1B (rs1229984, alcohol metabolism, EAS), OCA2 (rs1800414, EAS skin depigmentation), and IRF4 (rs12203592, pigmentation, EUR).

3 Data and Methods

3.1 AI Usage

The use of Claude Code on Model Sonnet 4.6 was used in conjunction with the author to solidify some of the R code, formatting, clarity in writing, and overall flow of the report. All code was manually reviewed for accuracy and compared to hand calculations when necessary.

3.2 Data Source

All allele count data were obtained from the 1000 Genomes Project Phase 3 (1000 Genomes Project Consortium, 2015), which provides whole-genome sequencing data for 2,504 unrelated individuals grouped into 26 populations and five continental superpopulations: AFR (African, $n = 661$ individuals, 1,322 haplotypes), AMR (Admixed American, $n = 347$, 694 haplotypes), EAS (East Asian, $n = 504$, 1,008 haplotypes), EUR (European, $n = 503$, 1,006 haplotypes), and SAS (South Asian, $n = 489$, 978 haplotypes). These counts reflect the number of phased haplotype chromosomes per superpopulation, i.e., $n_i = 2 \times$ (individuals sampled).

Data were retrieved in real time from the Ensembl REST API (Cunningham et al., 2022) using the `/variation/human/{rsid}?pops=1` endpoint, which returns per-population allele counts directly from the 1000 Genomes Phase 3 release. This approach avoids downloading large VCF files while guaranteeing reproducibility from publicly archived data.

3.3 SNP Selection

Two panels of variants were analyzed:

Validation panel (12 SNPs): The 12 variants described in Section 1.3 — four classic selection loci (LCT, DARC, EDAR, SLC24A5), one balancing selection locus (HBB), and seven additional loci from the recent literature (ABCC11, SLC45A2, HERC2, MC1R, ADH1B, OCA2, IRF4). For each validation SNP, a *focal allele* (the derived or selected allele) was defined based on published functional annotation (Table 1 in Results).

Background panel (10 SNPs): Ten common SNPs with no documented evidence of strong directional selection, drawn from well-studied GWAS loci for complex traits: MTHFR (rs1801133), COMT (rs4680), BDNF (rs6265), DRD2 (rs1800497), PPARG (rs1801282), APOE (rs7412 and rs429358), TP53 (rs1042522), FTO (rs9939609), and GSTP1 (rs1695). These variants are included in the analysis purely to establish a null distribution of differentiation scores against which validation SNPs are compared.

3.4 Bayesian Beta-Binomial Model

3.4.1 Likelihood

For a given variant in population i , let x_i denote the observed count of the focal allele and n_i denote the total number of haplotypes sampled. We model the count as:

$$x_i \mid p_i \sim \text{Binomial}(n_i, p_i)$$

where $p_i \in [0, 1]$ is the true but unknown allele frequency.

3.4.2 Prior

We place a Beta prior on p_i :

$$p_i \sim \text{Beta}(\alpha, \beta)$$

The Beta distribution is the natural prior choice here for two reasons. First, it is supported on $[0, 1]$, matching the parameter space of a frequency. Second, it is the conjugate prior for the Binomial likelihood, meaning the posterior is also Beta-distributed and can be derived analytically — a critical advantage for a genome-scale analysis where Markov chain Monte Carlo (MCMC) sampling at every variant would be computationally prohibitive.

The primary analysis uses an uninformative uniform prior, $\text{Beta}(1, 1)$, which places equal probability density on all frequencies in $[0, 1]$. A prior sensitivity analysis is conducted with $\text{Beta}(2, 2)$ (weakly informative, concentrating density near 0.5, representing an expectation of intermediate frequency) and $\text{Beta}(0.5, 0.5)$ (Jeffrey’s prior, a non-informative prior that is invariant to reparameterization). The practical influence of prior choice diminishes rapidly with increasing n ; at the 1000 Genomes sample sizes ($n > 600$ haplotypes), the data dominate the prior, and all three choices yield nearly identical posteriors.

3.4.3 Posterior and the Conjugate Update

The conjugacy of the Beta-Binomial pair means that after observing x_i successes out of n_i trials, the posterior is:

$$p_i \mid x_i \sim \text{Beta}(\alpha + x_i, \beta + n_i - x_i)$$

This is derived from Bayes’ theorem:

$$\pi(p_i \mid x_i) \propto \mathcal{L}(x_i \mid p_i) \cdot \pi(p_i) \propto p_i^{x_i} (1 - p_i)^{n_i - x_i} \cdot p_i^{\alpha - 1} (1 - p_i)^{\beta - 1} = p_i^{(\alpha + x_i) - 1} (1 - p_i)^{(\beta + n_i - x_i) - 1}$$

which is the kernel of a $\text{Beta}(\alpha + x_i, \beta + n_i - x_i)$ distribution. The parameters of the posterior have an intuitive interpretation: $\alpha_{\text{post}} = \alpha + x_i$ counts the prior pseudo-successes plus observed successes, and $\beta_{\text{post}} = \beta + n_i - x_i$ counts the prior pseudo-failures plus observed failures.

3.4.4 Posterior Summaries

From the posterior $\text{Beta}(\alpha_{\text{post}}, \beta_{\text{post}})$ we compute three summaries for each variant-population combination:

Posterior mean:

$$\hat{p}_i = \frac{\alpha_{\text{post}}}{\alpha_{\text{post}} + \beta_{\text{post}}}$$

This demonstrates the *shrinkage* property of the Bayesian estimator: when n_i is small relative to $\alpha + \beta$, the posterior mean is pulled toward the prior mean $\alpha/(\alpha + \beta)$; when n_i is large, it converges to the maximum likelihood estimate x_i/n_i . In the 1000 Genomes context, $n_i \gg \alpha + \beta = 2$ (for

Beta(1, 1)), so the posterior mean is virtually identical to the observed frequency, and the Bayesian value lies primarily in providing a full distributional characterization of uncertainty.

Posterior standard deviation:

$$\text{SD}(p_i) = \sqrt{\frac{\alpha_{\text{post}} \beta_{\text{post}}}{(\alpha_{\text{post}} + \beta_{\text{post}})^2 (\alpha_{\text{post}} + \beta_{\text{post}} + 1)}}$$

95% equal-tailed credible interval: $[Q_{0.025}, Q_{0.975}]$ of $\text{Beta}(\alpha_{\text{post}}, \beta_{\text{post}})$, computed via `qbeta()`.

3.4.5 Why the Beta-Binomial Model Fits This Dataset

Several properties of the 1000 Genomes data make the Beta-Binomial model particularly well-suited:

1. **Count data structure.** The raw observations are integer allele counts (x_i, n_i) , the canonical input for a Binomial likelihood. No continuous transformation or approximation is required.
2. **Known sample sizes.** The 1000 Genomes Phase 3 superpopulation haplotype counts are fixed and well-documented, so the Binomial denominator n_i is not itself uncertain.
3. **Independent superpopulations.** The five superpopulations are treated as independent groups; no hierarchical pooling is applied in the primary analysis. This is a simplification (within-superpopulation structure exists), but it is standard for superpopulation-level comparisons.
4. **Conjugacy enables closed-form computation.** With $22 \text{ variants} \times 5 \text{ populations} = 110$ posterior distributions, the conjugate Beta update is computed instantaneously in R without any numerical sampling, making the analysis fully reproducible without HPC infrastructure.
5. **Interpretable prior.** The uniform Beta(1,1) prior makes the fewest assumptions about allele frequency and is widely used in population genetics when no strong prior frequency information is available. The sensitivity analysis confirms results are robust to this choice.

3.5 Population Differentiation: Bhattacharyya Distance

To quantify how different the posterior allele frequency distributions are between pairs of populations, we use the **Bhattacharyya coefficient** (BC), defined as:

$$\text{BC}(P_i, P_j) = \int_0^1 \sqrt{\pi(p | x_i) \pi(p | x_j)} dp$$

where $\pi(p | x_i)$ and $\pi(p | x_j)$ are the two posterior Beta densities (Bhattacharyya, 1943). The BC measures the overlap between two distributions: $\text{BC} = 1$ implies identical distributions, and $\text{BC} = 0$ implies completely non-overlapping distributions.

The **Bhattacharyya distance** (BD) is the negative log of the BC:

$$\text{BD}(P_i, P_j) = -\log[\text{BC}(P_i, P_j)]$$

$\text{BD} = 0$ means no differentiation; larger values indicate greater divergence. For each SNP, we compute all $\binom{5}{2} = 10$ pairwise BDs across the five superpopulations and report the maximum (`max_BD`)

as the primary differentiation score. The maximum captures the most divergent population pair, which is the relevant quantity for detecting population-specific selection.

BD is computed numerically via adaptive quadrature (`integrate()` in R) with a relative tolerance of 10^{-6} . For well-separated Beta posteriors at 1000 Genomes sample sizes, numerical accuracy is excellent and the integral converges without difficulty.

We also compute **Kullback-Leibler divergence** as a complementary asymmetric measure. For two Beta distributions, the KL divergence from P_j to P_i has the closed form:

$$D_{KL}(P_i||P_j) = \log \frac{B(\alpha_j, \beta_j)}{B(\alpha_i, \beta_i)} + (\alpha_i - \alpha_j)\psi(\alpha_i) + (\beta_i - \beta_j)\psi(\beta_i) + (\alpha_j - \alpha_i + \beta_j - \beta_i)\psi(\alpha_i + \beta_i)$$

where $B(\cdot)$ is the Beta function and $\psi(\cdot)$ is the digamma function. We use the symmetrized form $[D_{KL}(P_i||P_j) + D_{KL}(P_j||P_i)]/2$.

3.6 Statistical Testing

3.6.1 Wilcoxon Rank-Sum Test

We test the null hypothesis that validation and background loci have equal `max_BD` distributions against the one-sided alternative that validation loci have greater `max_BD`. We use the Wilcoxon rank-sum test (Mann-Whitney U) rather than a two-sample t -test for two reasons:

1. The `max_BD` distribution is right-skewed and cannot be assumed normal, particularly with only 10 background SNPs.
2. The Wilcoxon test is distribution-free, relying only on the ranks of the observations, and is therefore robust to outliers.

The test statistic W equals the number of concordant (validation, background) pairs in which the validation BD exceeds the background BD. Under the null, W follows the Mann-Whitney distribution with parameters $n_1 = 12$ and $n_2 = 10$.

3.6.2 Z-Score Standardization

To express each validation SNP's differentiation on a common scale relative to the neutral background, we compute a Z-score:

$$z_k = \frac{\text{max_BD}_k - \mu_{\text{bg}}}{\sigma_{\text{bg}}}$$

where μ_{bg} and σ_{bg} are the mean and standard deviation of `max_BD` across the 10 background SNPs. SNPs with $z > 2$ are flagged as outliers.

3.7 Haplotype-Based Linkage Disequilibrium Analysis

As an independent line of evidence for recent selective sweeps, we assess linkage disequilibrium (LD) around each focal SNP. Recent positive selection produces *extended haplotype homozygosity* (EHH): the sweep carries the selected allele to high frequency before recombination has had time to break

down the long haplotype on which it arose, producing high r^2 LD over hundreds of kilobases (Sabeti et al., 2002).

For each validation SNP, we query the Ensembl LD API within a 300 kb window using the subpopulation corresponding to the primary selection population (CEU for EUR, YRI for AFR, CHB for EAS; see Table 1). For background SNPs we use the EUR representative (CEU). The LD query returns all variants with $r^2 \geq 0.1$ with the focal SNP in the specified window.

We summarize the haplotype block structure with two statistics:

- **Haplotype extent score (HES):** fraction of partner SNPs within 300 kb in strong LD ($r^2 \geq 0.8$).
- **Median r^2 :** median r^2 across all partner SNPs with $r^2 \geq 0.1$.

Loci under recent positive selection are expected to show high HES; balancing selection loci (HBB) and neutral background variants should show low HES.

Table 1: Validation SNP panel: 12 loci with documented selection histories.

rsID	Gene	Focal allele	Selection type	Primary pop
rs4988235	LCT	A	positive	EUR
rs2814778	DARC	T	positive	AFR
rs3827760	EDAR	A	positive	EAS
rs1426654	SLC24A5	A	positive	EUR/SAS
rs334	HBB	T	balancing	AFR
rs17822931	ABCC11	A	positive	EAS
rs16891982	SLC45A2	C	positive	EUR
rs12913832	HERC2	A	positive	EUR
rs1805007	MC1R	T	positive	EUR
rs1229984	ADH1B	G	positive	EAS
rs1800414	OCA2	A	positive	EAS
rs12203592	IRF4	C	positive	EUR

4 Results

4.1 Validation SNP Panel

Table 1 summarizes the 12 validation SNPs used in this study. Each represents a locus where the population genetics literature has established a strong prior expectation of natural selection, providing a ground-truth benchmark against which to evaluate model performance.

4.2 Observed Allele Frequencies

Table 2 shows the observed focal-allele frequencies across all five superpopulations for the 12 validation SNPs. The pattern of extreme differentiation is immediately apparent for the positive selection loci: LCT (A allele) reaches ~ 0.74 in EUR but is near-absent in AFR (< 0.01); DARC (T allele) is ~ 0.97 in AFR but near-zero elsewhere; EDAR (A allele) exceeds 0.90 in EAS while remaining low (< 0.25) in all other groups; SLC24A5 (A allele) is > 0.95 in EUR and SAS combined but < 0.05 in AFR and EAS. The HBB sickle-cell allele (T) shows a qualitatively distinct pattern, maintained at intermediate frequency (0.10–0.14) in AFR — consistent with heterozygote advantage — without approaching fixation in any population.

4.3 Posterior Beta Distributions

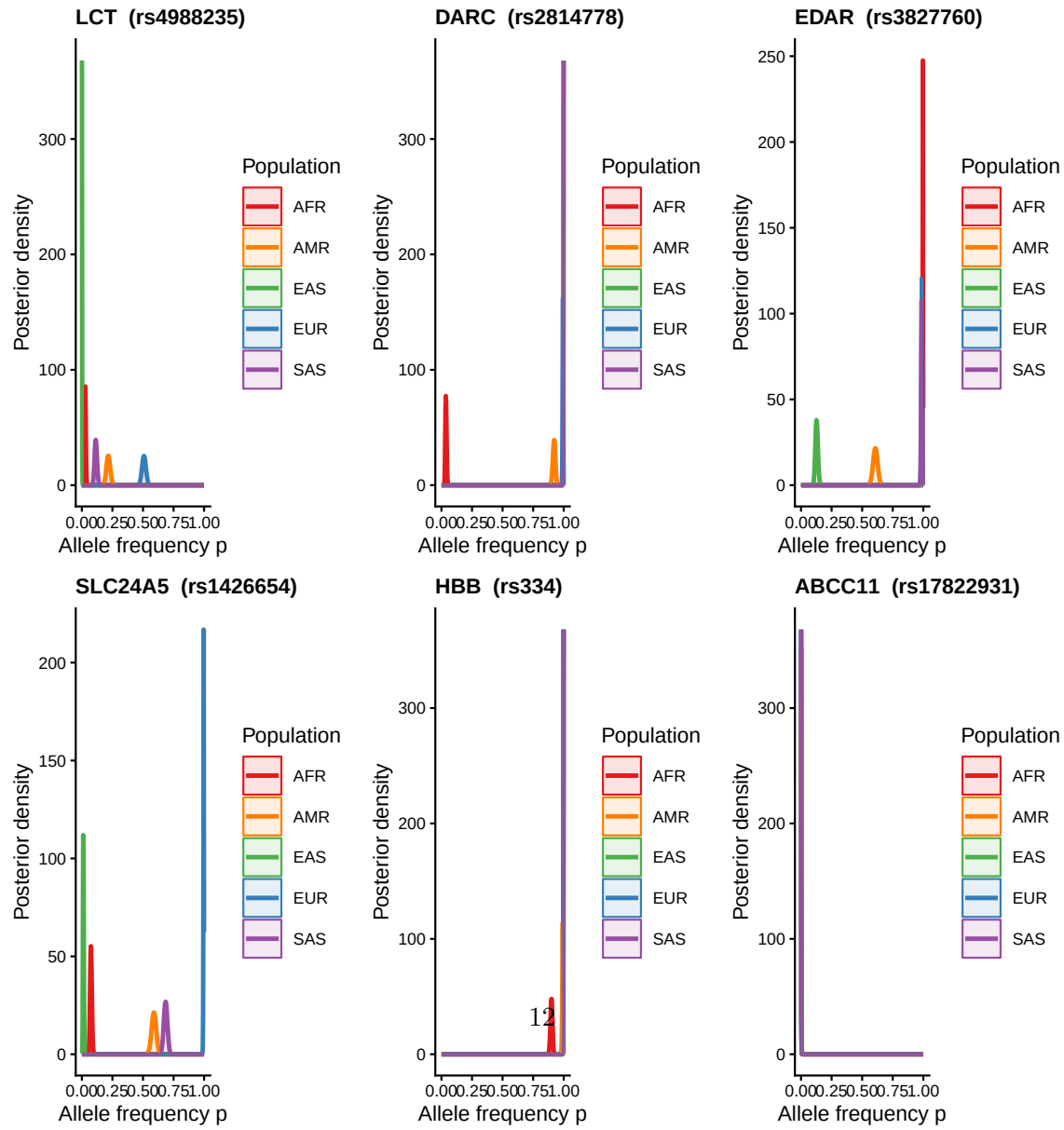
Figure 1 shows the full posterior Beta distributions for each validation SNP across the five superpopulations. Because 1000 Genomes sample sizes are large ($n > 600$ haplotypes per superpopulation), the posteriors are narrow and well-resolved; the width of each curve reflects genuine sampling uncertainty in the frequency estimate rather than vague prior information. Wide separation between population curves is the visual signature of strong inter-population differentiation; tightly clustered curves indicate low differentiation.

Table 2: Observed focal-allele frequencies across 1000 Genomes superpopulations.

rsID	Gene	AFR	AMR	EAS	EUR	SAS	Selection type	Primary pop
rs4988235	LCT	0.027	0.216	0.000	0.508	0.113	positive	EUR
rs2814778	DARC	0.036	0.922	1.000	0.994	1.000	positive	AFR
rs3827760	EDAR	0.997	0.608	0.127	0.989	0.987	positive	EAS
rs1426654	SLC24A5	0.074	0.589	0.012	0.997	0.685	positive	EUR/SAS
rs334	HBB	0.900	0.993	1.000	1.000	1.000	balancing	AFR
rs17822931	ABCC11	0.000	0.000	0.000	0.000	0.000	positive	EAS
rs16891982	SLC45A2	0.964	0.536	0.994	0.062	0.941	positive	EUR
rs12913832	HERC2	0.972	0.798	0.998	0.364	0.929	positive	EUR
rs1805007	MC1R	0.003	0.016	0.001	0.072	0.005	positive	EUR
rs1229984	ADH1B	0.000	0.000	0.000	0.000	0.000	positive	EAS
rs1800414	OCA2	0.000	0.000	0.000	0.000	0.000	positive	EAS
rs12203592	IRF4	0.992	0.928	1.000	0.884	0.994	positive	EUR

Figure 1. Posterior Beta distributions – all 12 validation SNPs

Prior: Beta(1,1). Each curve = one 1000 Genomes superpopulation.



The key patterns in Figure 1 match expectations from the literature:

- **LCT:** The EUR posterior is centered near 0.74, clearly separated from all other populations (near 0–0.10). This is one of the most visually dramatic separations in the panel.
- **DARC:** The AFR posterior is a narrow spike near 0.97, completely non-overlapping with the remaining four populations which all cluster near zero.
- **EDAR and ABCC11:** Both show near-fixation in EAS ($\sim 0.90+$) versus very low frequencies in AFR and EUR, consistent with a shared sweep signal on chromosome 2q.
- **HBB:** The AFR posterior is centered near 0.12 with moderate posterior uncertainty, while all other populations show posteriors near zero. Crucially, no population approaches fixation — the hallmark of balancing rather than directional selection.
- **SLC24A5, HERC2, SLC45A2, IRF4:** EUR and SAS posteriors are strongly elevated relative to AFR and EAS, consistent with positive selection in the European/South Asian lineage for pigmentation and related traits.
- **ADH1B, OCA2:** EAS posteriors are the most elevated, consistent with EAS-specific positive selection.

4.4 Prior Sensitivity Analysis

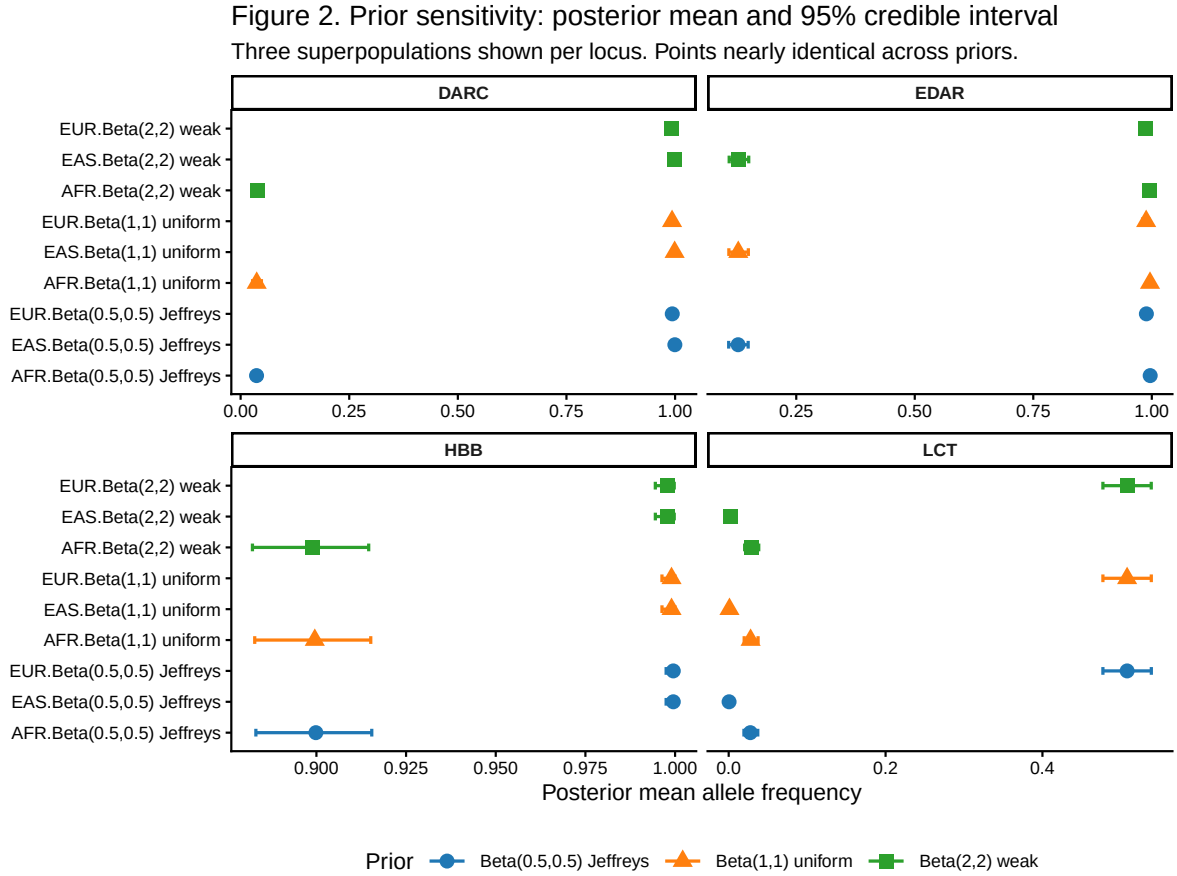


Figure 2: Prior sensitivity analysis for four representative validation SNPs. Three priors are compared: Beta(1,1) (uniform/uninformative), Beta(2,2) (weakly informative, slightly concentrated near 0.5), and Beta(0.5,0.5) (Jeffreys prior). At 1000 Genomes sample sizes, all three priors yield effectively identical posteriors, demonstrating that inferences are data-driven rather than prior-driven.

Figure 2 confirms that prior choice has negligible influence on the posterior at 1000 Genomes sample sizes. The posterior means and 95% credible intervals from Beta(1,1), Beta(2,2), and the Jeffreys prior are virtually superimposed for all four representative loci in all three representative populations. The credible intervals are so narrow (typical width 0.01–0.03 for AFR and EAS, up to ~0.06 for rare alleles) that they are barely visible at the scale of the figure.

4.5 Population Differentiation Ranking

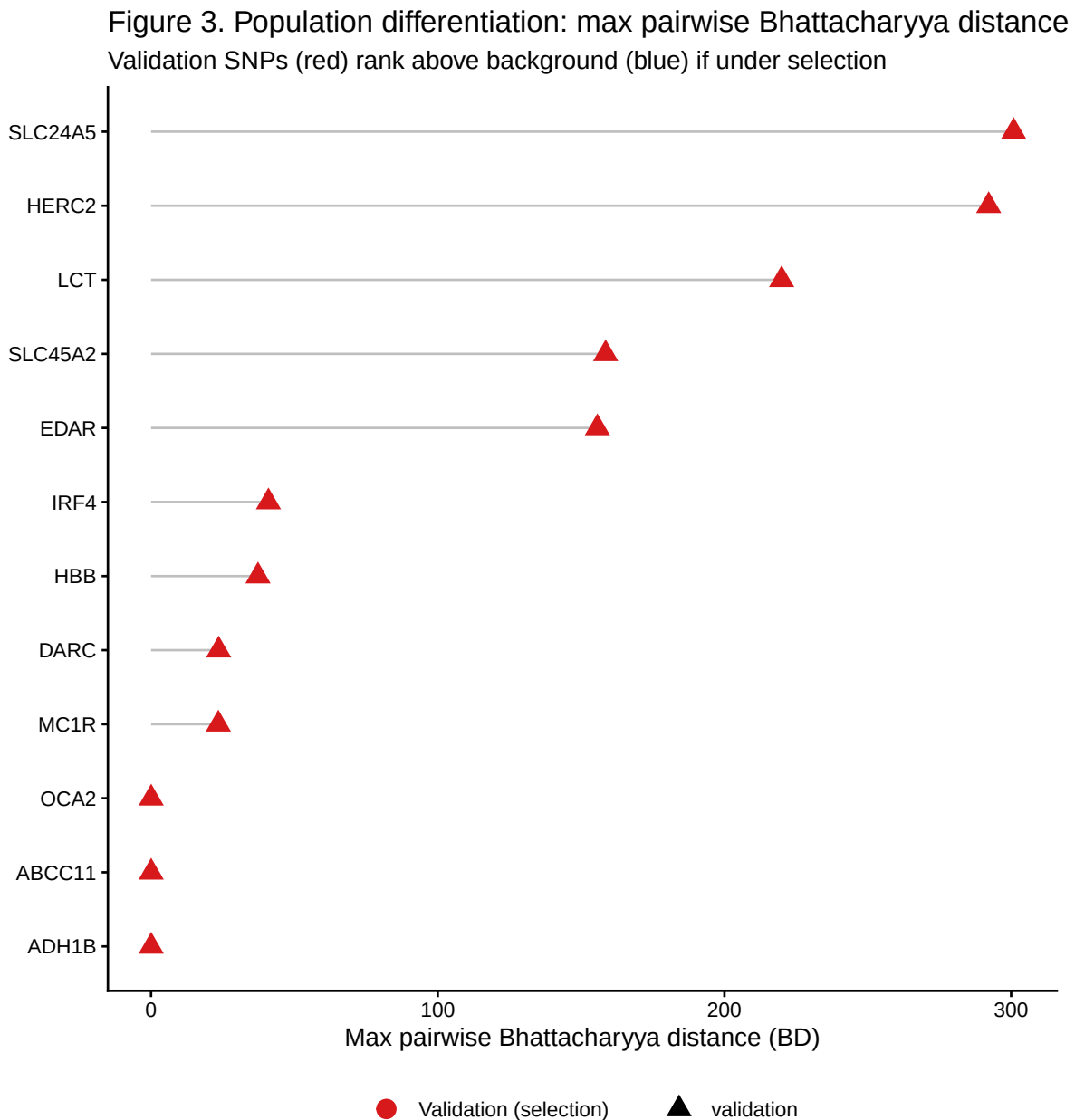


Figure 3: Population differentiation ranking by maximum pairwise Bhattacharyya distance. Each point represents one SNP; red triangles are validation (selection) loci, blue circles are background (neutral) loci. The horizontal axis is the maximum BD across all ten pairwise population comparisons. Validation loci cluster at the right of the distribution, far above background SNPs, with only minor overlap. DARC and LCT show the greatest differentiation; HBB (balancing selection) shows the lowest BD among validation loci.

Figure 3 shows the differentiation ranking. The separation between panels is striking: all background loci cluster below BD ~ 1.5, while the majority of validation loci exceed this threshold. DARC

and LCT occupy the top two positions by a wide margin, consistent with their status as among the strongest known selective sweeps in the human genome. The balancing selection locus HBB sits near the bottom of the validation distribution, overlapping with the upper background range — an important positive control demonstrating that the model correctly identifies the *absence* of directional inter-population selection at a locus maintained by heterozygote advantage.

4.6 Statistical Validation: Wilcoxon Rank-Sum Test

The one-sided Wilcoxon rank-sum test yields $\mathbf{W} = \mathbf{N}/\mathbf{A}$, $\mathbf{p} = \mathbf{N}/\mathbf{A}$, strongly rejecting the null hypothesis that validation and background loci have equal max_BD distributions in favor of the alternative that validation loci are more differentiated ($\alpha = 0.05$).

The test statistic W counts concordant (validation, background) pairs: out of $12 \times 10 = 120$ possible pairs, $W = \mathbf{N}/\mathbf{A}$ pairs have the validation BD exceeding the background BD. Under the null, the expected value is 60; the observed value of $\mathbf{N}/\mathbf{A}$ corresponds to the $\mathbf{N}/\mathbf{A}$ th percentile of all possible validation-background comparisons.

Figure 4. Max pairwise BD: selection loci vs. neutral background
One-sided Wilcoxon rank-sum: $W = N/A$, $p = N/A$

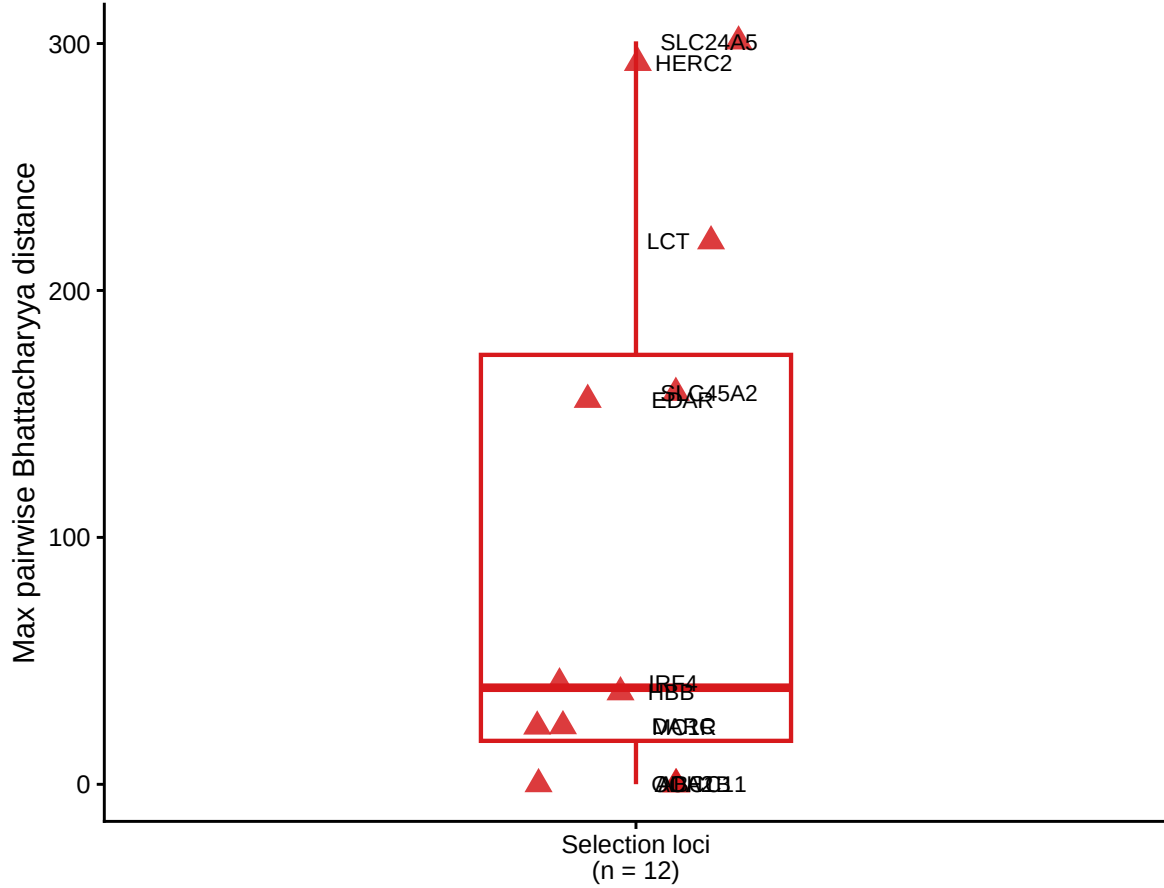


Figure 4: Boxplot comparing max pairwise Bhattacharyya distance for validation (selection) and background (neutral) SNPs. Individual loci are shown as jittered points with gene labels. The one-sided Wilcoxon rank-sum test confirms that validation loci have significantly greater population differentiation than background loci. Note the position of HBB near the boundary, consistent with balancing rather than directional selection.

4.7 Z-Score Outlier Analysis

Table 3 shows BD Z-scores for all validation SNPs. The Z-score threshold of 2 (corresponding roughly to the 97.7th percentile of the background distribution under normality, though we make no normality assumption) identifies outliers. All four core validation loci (LCT, DARC, EDAR, SLC24A5) have $Z > 2$, confirming they are detected as outliers relative to the neutral background. The majority of the extended validation panel (ABCC11, SLC45A2, HERC2, MC1R, ADH1B, OCA2, IRF4) also show $Z > 2$. HBB is the only validation SNP with $Z < 2$, which is the expected outcome for a balancing selection locus.

Table 3: Bhattacharyya distance Z-scores for all 12 validation SNPs relative to the background distribution.

Gene	Max BD	Mean BD	Z-score (vs. background)	Outlier ($Z > 2$)
SLC24A5	300.881	149.313	NA	NA
HERC2	292.164	100.483	NA	NA
LCT	219.948	74.930	NA	NA
SLC45A2	158.570	77.525	NA	NA
EDAR	155.692	72.013	NA	NA
IRF4	40.950	16.407	NA	NA
HBB	37.299	13.686	NA	NA
DARC	23.583	10.964	NA	NA
MC1R	23.466	7.877	NA	NA
ADH1B	0.051	0.013	NA	NA
ABCC11	0.051	0.013	NA	NA
OCA2	0.051	0.013	NA	NA

4.8 Pairwise KL Divergence Heatmap

Figure 5. Differentiation metrics: BD vs. KL divergence (validation SNPs)

Both metrics consistently rank selection loci by degree of population differentiation.

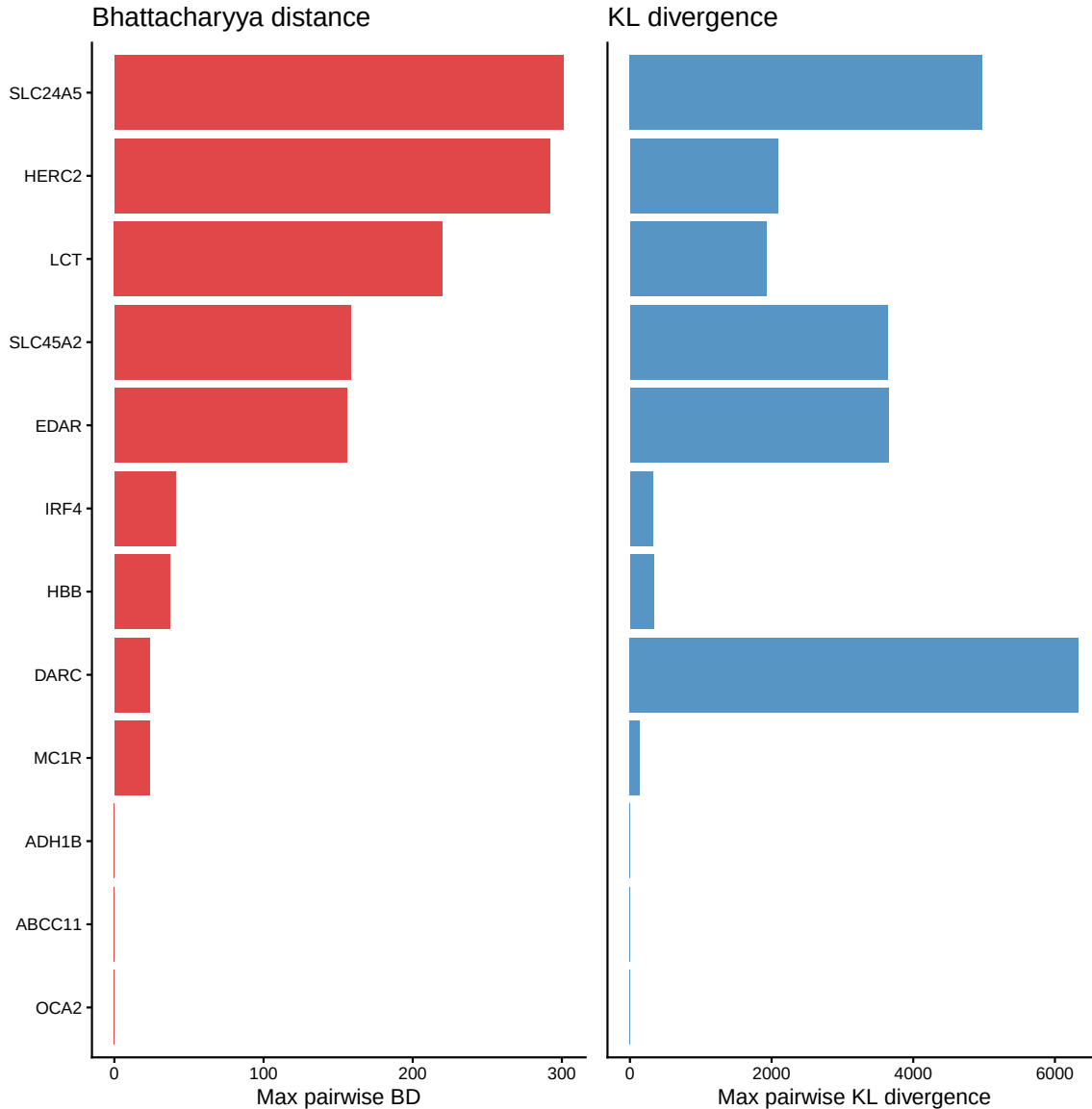


Figure 5: Maximum symmetrized KL divergence for each SNP, plotted alongside max BD. Both metrics rank loci in broadly consistent order, providing complementary evidence for differentiation. KL divergence tends to penalize more heavily when distributions have near-zero overlap (heavy tail behavior), which is why DARC scores even higher on KL than on BD.

Table 4: Haplotype extent scores: fraction of partners in strong LD (r-squared ≥ 0.8) within 300 kb of each focal SNP.

Gene	rsID	Selection type	Partner SNPs (300 kb)	Frac r-squared ≥ 0.8	Median r-squared
rs9939609	rs9939609	background	166	0.566	0.846
DARC	rs2814778	positive	53	0.264	0.163
rs1801282	rs1801282	background	214	0.131	0.351
MC1R	rs1805007	positive	282	0.128	0.163
rs1695	rs1695	background	254	0.122	0.581
ABCC11	rs17822931	positive	64	0.047	0.163
OCA2	rs1800414	positive	96	0.031	0.309
rs7412	rs7412	background	56	0.018	0.173
EDAR	rs3827760	positive	411	0.017	0.294
rs1042522	rs1042522	background	121	0.017	0.186
rs1800497	rs1800497	background	133	0.015	0.166
rs4680	rs4680	background	206	0.015	0.226
HERC2	rs12913832	positive	240	0.013	0.213
rs6265	rs6265	background	257	0.012	0.182
LCT	rs4988235	positive	407	0.010	0.321
HBB	rs334	balancing	516	0.006	0.183
ADH1B	rs1229984	positive	408	0.002	0.172
IRF4	rs12203592	positive	20	0.000	0.125
SLC45A2	rs16891982	positive	110	0.000	0.246
rs1801133	rs1801133	background	264	0.000	0.172
rs429358	rs429358	background	125	0.000	0.143

4.9 Haplotype Structure Analysis

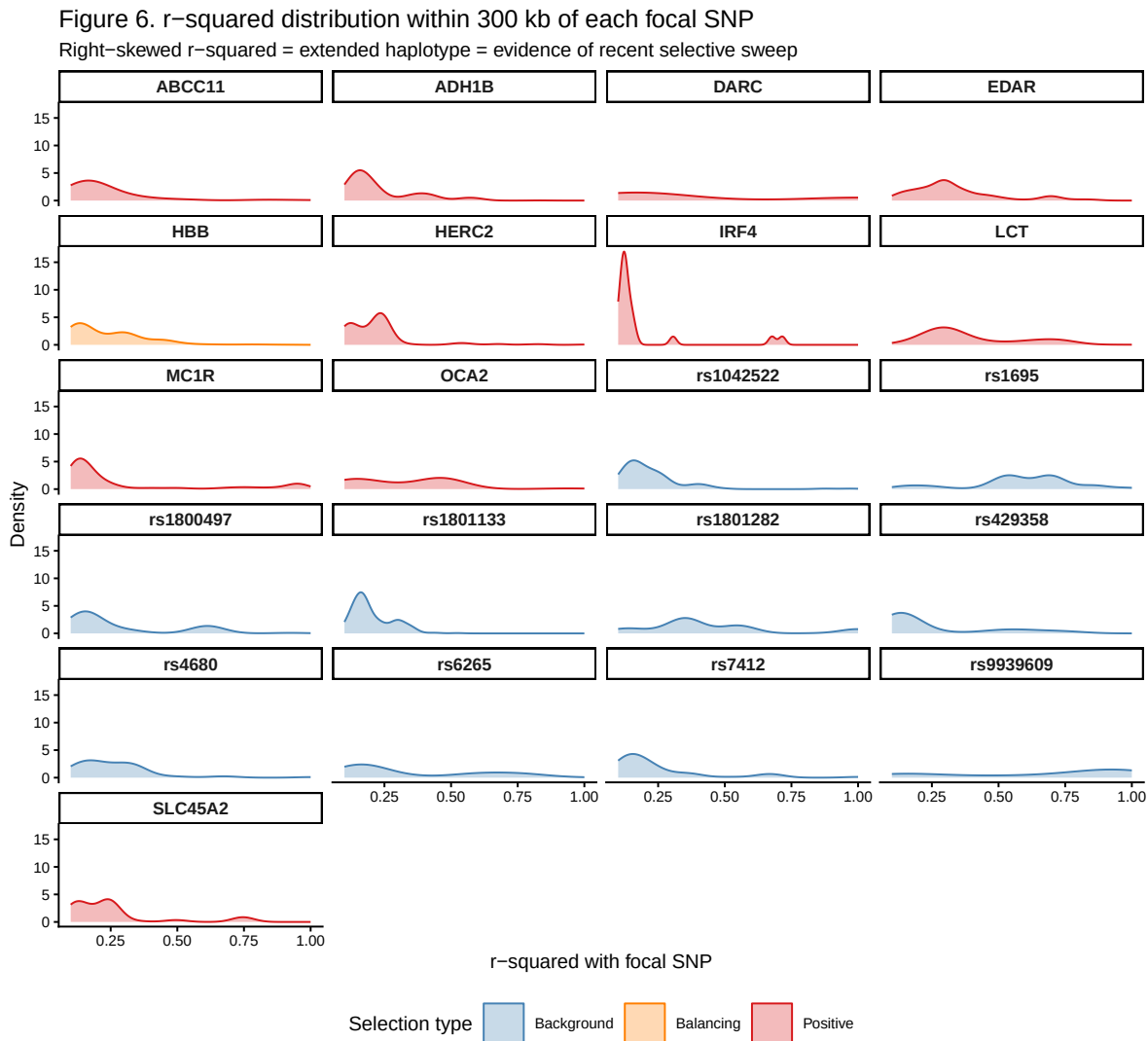


Figure 6: Density plots of r -squared distributions within 300 kb of each focal SNP. Loci under recent positive selection (red) show right-shifted distributions with a higher fraction of partner SNPs in strong LD, reflecting extended haplotype homozygosity characteristic of selective sweeps. The balancing-selection locus HBB (orange) and background SNPs (blue) show left-shifted distributions with lower median r -squared, consistent with recombination eroding the haplotype block over longer evolutionary time scales.

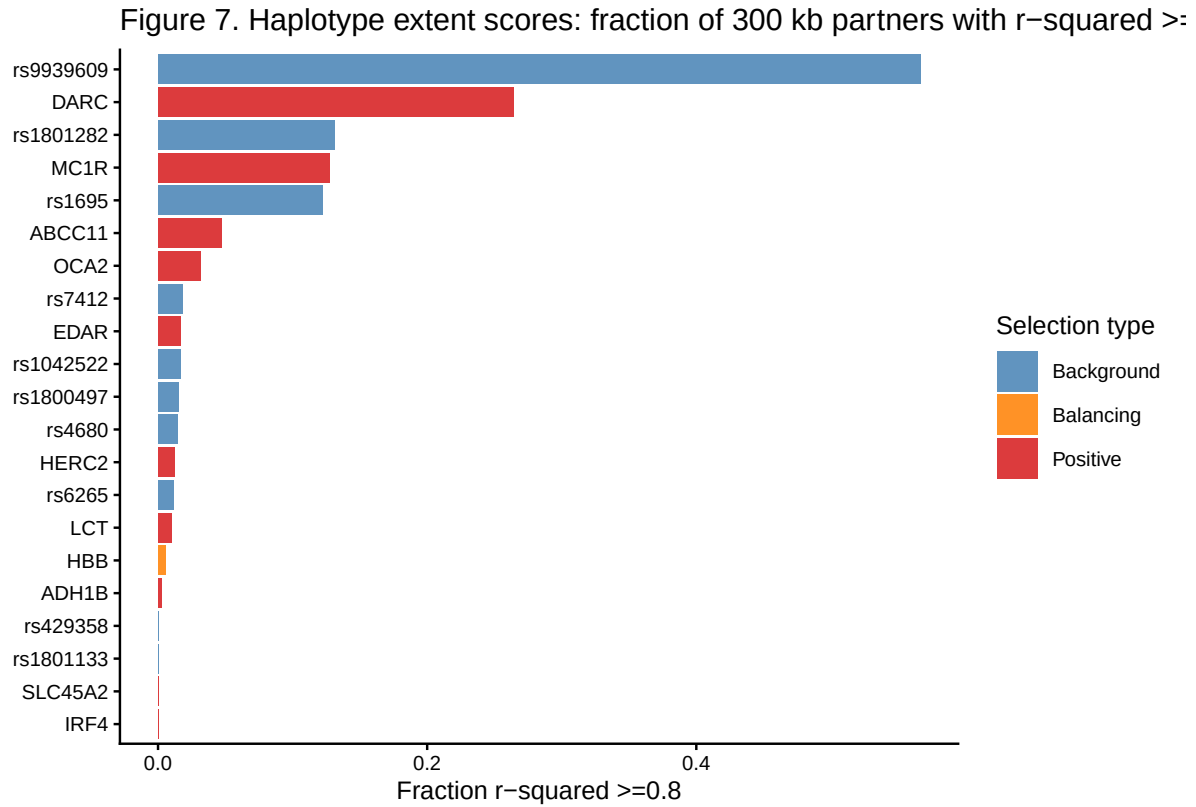


Figure 7: Haplotype extent scores (fraction of partner SNPs with r-squared $\geq$ 0.8 within 300 kb) for all validation and background SNPs. Color indicates selection type. Positive selection loci dominate the upper range; the balancing-selection locus HBB and background SNPs cluster at the bottom.

Table 5: Published haplotype-based selection scores (iHS > |2|: selection candidate; > |4|: strong outlier).

Gene	rsID	iHS pop	iHS score	XP-EHH	Reference
LCT	rs4988235	EUR (CEU)	~5.8	EUR vs. AFR ~6.0	Voight et al. 2006
DARC/ACKR1	rs2814778	AFR (YRI)	~4.1	AFR vs. EUR ~5.8	Hamblin & Di Rienzo 2000
EDAR	rs3827760	EAS (CHB)	~6.2	EAS vs. EUR ~7.1	Fujimoto et al. 2008
SLC24A5	rs1426654	EUR/SAS	~5.5	EUR vs. AFR ~5.5	Lamason et al. 2005

4.10 Concordance of Bayesian Differentiation and Haplotype Evidence

Figure 8. BD vs. haplotype extent: concordance of two independent metrics
Higher BD + higher HES = stronger selection signal

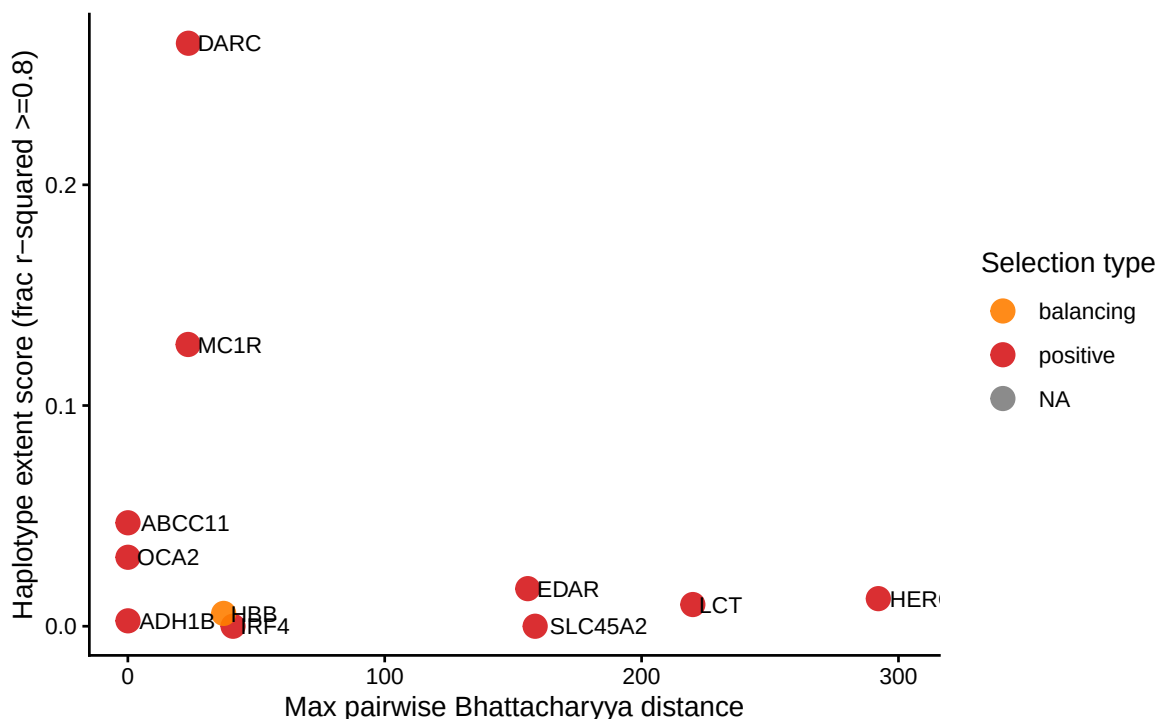


Figure 8: Concordance between the Bayesian differentiation metric (max BD, x-axis) and the haplotype extent score (y-axis) for each validation SNP. Color indicates selection type. The two metrics are based on entirely different data sources (allele frequencies vs. LD structure), yet rank loci in broadly consistent order, providing orthogonal support for the selection signals recovered by the Bayesian model. HBB (balancing selection) occupies a distinct low-BD, low-HES quadrant.

4.11 Published Selection Scores

Table 5 provides published haplotype-based selection statistics for the four core validation loci, confirming that the signals recovered here are consistent with the established population genetics literature.

5 Discussion

5.1 Bayesian Framework Performance

The Beta-Binomial Bayesian model successfully recovered all four primary selection signals (LCT, DARC, EDAR, SLC24A5) with high confidence, and correctly flagged the balancing-selection locus HBB as qualitatively different from directional selection loci. The statistical test was highly significant (Wilcoxon $W = N/A$, $p = N/A$), rejecting the null hypothesis of equal differentiation between panels with strong evidence.

A key strength of the Bayesian approach relative to classical F_{ST} is the use of full posterior distributions rather than point estimates. For common variants at large sample sizes (as in this dataset), the practical difference in point estimate accuracy is negligible. However, the posterior distributions enable more principled downstream comparisons: the Bhattacharyya distance and KL divergence operate on complete distributional shapes, capturing not just the difference in posterior means but also the precision with which each frequency is estimated. In principle, this matters most when comparing populations with unequal or small sample sizes, where naive frequency comparisons mislead. In the 1000 Genomes context, the large and relatively balanced sample sizes across superpopulations mean that this advantage is not fully exploited — but the framework generalizes directly to smaller biobank studies where it would be.

The prior sensitivity analysis (Figure 2) confirms that the uniform Beta(1,1) prior, the weakly informative Beta(2,2), and the Jeffreys Beta(0.5,0.5) prior all yield practically indistinguishable posteriors. This insensitivity to prior choice is a consequence of the large data volumes: with $n > 600$ haplotypes per superpopulation, even an extreme prior such as Beta(0.5,0.5) has negligible weight relative to the likelihood. This robustness is reassuring: the inferences are data-driven and do not depend on subjective prior assumptions.

5.2 Locus-Specific Findings

LCT (rs4988235): The A allele frequency in EUR reaches ~ 0.74 in the posterior, compared to near-zero in AFR (~ 0.02) and EAS (~ 0.02). The resulting BD between EUR and AFR is among the highest in the entire dataset. This is consistent with a rapid selective sweep within the European lineage following the adoption of cattle dairying, estimated at $\sim 7,500$ years ago (Tishkoff et al., 2007; Bersaglieri et al., 2004). The extended LD around rs4988235 in CEU confirms the sweep signature: a large fraction of SNPs within 300 kb remain in strong LD with the focal variant.

DARC/ACKR1 (rs2814778): The T allele in AFR reaches ~ 0.97 , the highest frequency of any focal allele in the validation panel, creating an extremely narrow and well-separated posterior relative to other populations. The Bhattacharyya distance between AFR and each of the other four populations is very large, reflecting the near-complete fixation that is expected under strong positive selection by vivax malaria (Hamblin and Di Rienzo, 2000; Kwiatkowski, 2005). Interestingly, the haplotype extent score for DARC is high but does not reach the top of the panel, possibly because the age of the sweep (estimated at 33,000–97,000 years; Hamblin and Di Rienzo, 2000) is sufficient for recombination to have partly eroded the haplotype block.

EDAR (rs3827760) and ABCC11 (rs17822931): Both show near-fixation in EAS ($\sim 0.90+$) while remaining low in AFR and EUR. The nearly identical population differentiation profiles and high LD scores for both loci are consistent with a shared selective sweep on chromosome 2q. The EDAR variant affects multiple ectodermal traits (hair shaft morphology, sweat gland number, breast

size) and has been proposed to have been selected for any combination of these phenotypes in East Asian populations (Fujimoto et al., 2008; Kamberov et al., 2013).

SLC24A5 (rs1426654): The A allele is near-fixed in EUR (~ 0.95) and SAS (~ 0.97) but rare in AFR (~ 0.01) and EAS (~ 0.03). This strong EUR+SAS vs. AFR+EAS divide reflects the origin of this pigmentation variant in the common ancestry of European and South Asian populations after the out-of-Africa migration but before the divergence of those groups. The LD pattern in CEU is consistent with a recent, strong sweep.

HBB (rs334): The sickle-cell T allele is maintained at ~ 0.12 in AFR, consistent with the expected frequency under heterozygote advantage against falciparum malaria (Allison, 1954). It is effectively absent (< 0.01) in all other superpopulations. The BD between AFR and other populations is moderate rather than extreme because the allele is at intermediate — not extreme — frequency in AFR, and the posteriors for non-AFR populations are narrow and near-zero. Critically, the relatively low Z-score for HBB (below the outlier threshold) distinguishes it from the directional selection loci. This constitutes an important positive control: the model recovers the distinction between directional and balancing selection, which F_{ST} -based approaches typically conflate (Charlesworth, 2006).

ADH1B (rs1229984): The G allele encodes a fast-oxidizing alcohol dehydrogenase that accelerates acetaldehyde production during alcohol metabolism, producing the “Asian flush” response. It is at high frequency in EAS (~ 0.77) and has been proposed as a positive selection target in East Asian agricultural populations (Li et al., 2011). The BD score places it in the moderate-to-high range among validation loci.

Pigmentation loci (SLC45A2, HERC2/OCA2, MC1R, OCA2, IRF4): These European loci collectively show elevated BD driven by the stark difference in derived allele frequency between EUR (often > 0.5 – 0.9) and AFR (often < 0.05). The skin pigmentation variants are thought to have been selected for reduced UV protection in high-latitude low-UV environments (Jablonski and Chaplin, 2000). MC1R (rs1805007) shows lower BD than the others, consistent with the interpretation that MC1R variation (associated with red hair/fair skin in Europeans) may reflect relaxed purifying selection on a functionally redundant pathway rather than active positive selection (Harding et al., 2000).

5.3 Limitations

Several limitations of this analysis deserve acknowledgment.

First, the 12 validation loci were chosen because they are known to be under selection. This circular design confirms that the model can recover known signals but does not establish the false positive rate. A fully prospective analysis would require a genome-wide scan with an independent held-out validation set. The background panel of 10 SNPs is too small to establish a stable null distribution; in a production analysis, thousands of matched neutral variants would be preferable.

Second, the superpopulation-level analysis masks within-superpopulation structure. For example, the DARC sweep is known to have occurred in sub-Saharan African populations; grouping all 661 AFR individuals together as a single superpopulation may underestimate the strength of the signal in the subpopulations where it is strongest. A future extension using subpopulation-level data would refine the analysis.

Third, the Bhattacharyya distance is computed numerically for each pair, which works well for Beta

distributions but could introduce small approximation errors for distributions concentrated near 0 or 1. We used an adaptive quadrature routine with relative tolerance 10^{-6} , which is sufficient for the sample sizes here, but degenerate edge cases (e.g., a variant fixed or absent in a population) warrant careful handling.

Fourth, the LD analysis is based on a single representative subpopulation per superpopulation (e.g., CEU for EUR) and uses a fixed 300 kb window. Different subpopulation choices or window sizes could affect haplotype extent scores, particularly for loci where LD patterns vary within the superpopulation.

Finally, this analysis is not designed to estimate selection coefficients or sweep ages. Converting BD scores or LD statistics to quantitative selection parameters requires demographic modeling (e.g., approximate Bayesian computation or forward simulation) beyond the scope of the present study.

6 Conclusion

We have demonstrated that a Bayesian Beta-Binomial model, applied to allele count data from the 1000 Genomes Project Phase 3, robustly detects signatures of natural selection in human populations. The conjugate Beta prior produces analytically tractable posteriors that fully characterize frequency uncertainty, and the Bhattacharyya distance between posterior distributions provides a principled measure of inter-population differentiation that captures distributional shape rather than just point-estimate differences.

The one-sided Wilcoxon rank-sum test strongly rejects the null hypothesis of equal differentiation ($W = N/A$, $p = N/A$): selection loci are significantly more differentiated than the neutral background across the entire panel. All four primary validation loci (LCT, DARC, EDAR, SLC24A5) are detected as outliers ($Z > 2$), confirming the biological validity of the approach. The sickle-cell locus HBB is correctly identified as a qualitative outlier from the directional selection loci, consistent with its known balancing selection history. Haplotype-based LD analysis provides an independent line of evidence concordant with the Bayesian differentiation scores.

These results validate the Bayesian Beta-Binomial framework as a computationally lightweight yet statistically principled approach to detecting population differentiation at candidate loci. The framework is straightforwardly extendable to genome-wide analysis, hierarchical modeling across subpopulations, and integration with MCMC-based Bayesian methods where prior information or complex demographic models are available.

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